

GitHub Actions

Why GitHub Actions

- GitHub actions are used to automate software workflows. It is CI/CD tool
- Automatically build your code
- No manual work is needed
- Reduce time
- No need to setup any servers.
- Cost effective

What is GitHub Actions

- GitHub Actions is a CI/CD (Continuous Integration and Continuous Deployment) tool built into GitHub that allows you to automate workflows directly from your repository.
- Automatically build your project when you push code
- Run tests on every pull request
- Run scheduled jobs

How it Works

GitHub actions have some components that are

- **Workflows:**
It uses workflows to automate the process. It will run on or more jobs at a time. Workflows are defined using YAML files in the repository. It will run when a triggered by an event in the repository. Workflows are declared under `.GitHub/workflows`
- **Event:**
It is a specific activity in a repository. It triggers a workflow run. Without an event, workflow will not run. Event tells the workflow when to start. Events are nothing but creates a pull request and pushes the repo.
- **Job:**
A job is a set of steps in a workflow. It is executed on the same runner. Each step will be executed in an ordered manner. Runs on a virtual machine. Can run multiple jobs in parallel.
- **Actions:**
An Action is a reusable unit of code. You can write your own actions, or you can find actions to use in your workflows in the GitHub Marketplace. It performs a specific task within a workflow.

- Runner:

A runner is server that runs the workflow when the workflows are triggered. Each runner can run on a single job at a time. Github provides some runners to run the workflows.